Name

Mathematics Advanced Year 11

Trigonometry with Acute Angles

Book 1

Version: 261021

Feedback:
https://MrDingMaths.com

Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-04 applies trigonometry to solve problems involving geometric shapes

Trigonometry with acute angles

- Use Pythagoras' theorem to find the exact sine, cosine and tangent ratios for angles of $30°$, $45°$ and $60°$
- Apply trigonometry to solve problems involving right-angled triangles in two dimensions, true and compass bearings, and angles of elevation and depression

Right-Angled Trigonometry

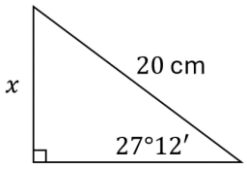
The Trigonometric Ratios

- There are three **trigonometric ratios: sine, cosine, and tangent,** which compare the lengths of two sides of a triangle.
- The ratios specify an angle, as the opposite and adjacent sides depend on the angle.
- You can remember the three ratios using SOH CAH TOA

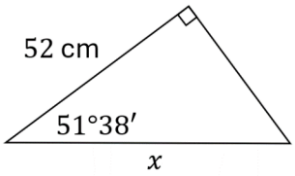
$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$	$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$	$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}}$

Example Calculate unknown sides when angle is in sexagesimal form

Calculate the value of x. Round your answer to 2 decimal places



$$\sin(27°12') = \frac{x}{20}$$
$$20\sin(27°12') = x$$
$$9.14\text{ cm} = x$$

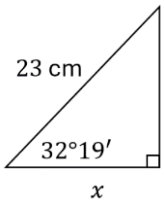


$$\cos(51°38') = \frac{52}{x}$$
$$x\cos(51°38') = 52$$
$$x = \frac{52}{\cos(51°38')}$$
$$= 84.78\text{ cm}$$

Guided Practice

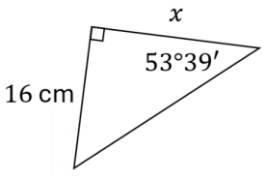
Calculate the value of x. Round to 2 decimal places.

a



19.44 cm

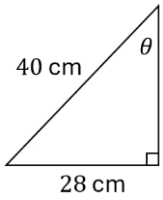
b



$x = 11.77$ cm

Example Calculate an unknown angle and round the result

Calculate the value of θ and round to the nearest minute.



$$\sin\theta = \frac{28}{40}$$

$$\theta = \sin^{-1}\left(\frac{28}{40}\right)$$

$$\theta \approx 44.4270$$

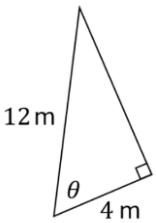
$$\approx 44° 25' 37.21''$$

$$\approx 44° 26'$$

Guided Practice

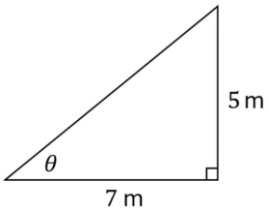
Calculate the value of θ and round to the nearest minute.

a



$\theta = 70°32'$

b



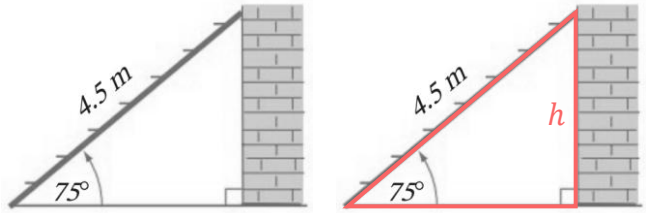
$\theta = 35°32'$

Problem-solving using Trigonometry

1. Identify the right-angled triangle.
2. Label the unknown side or angle x or θ, if it isn't already.
3. Solve for the unknown using trigonometry.

Example Solve practical problems involving trigonometry

Find how high the ladder reaches up the wall.



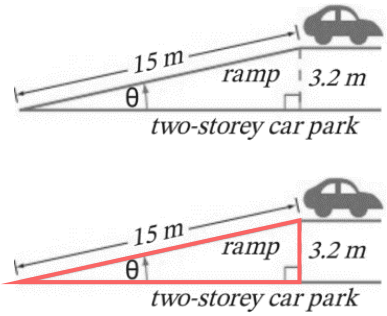
1. Triangle
2. Unknown (h)
3. Solve

$$\sin 75° = \frac{h}{4.5}$$

$$4.5 \sin 75° = h$$

$$x \approx 4.35 \text{ m}$$

Find the angle of the ramp.



1. Triangle
2. Unknown (θ)
3. Solve

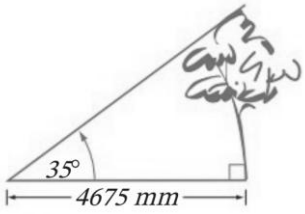
$$\sin \theta = \frac{3.2}{15}$$

$$\theta = \sin^{-1}\left(\frac{3.2}{15}\right)$$

$$\theta \approx 12.32°$$

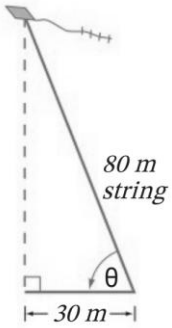
Guided Practice

a Determine the height of this tree.



$x = 3273.47$ mm

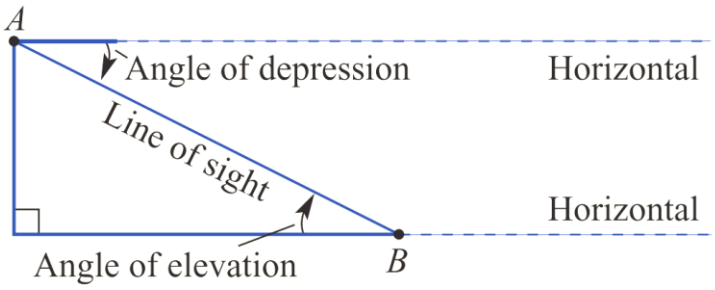
b Find the angle between the kite and the ground.



$\theta = 67.98°$

ⓘ Angles of Elevation and Depression

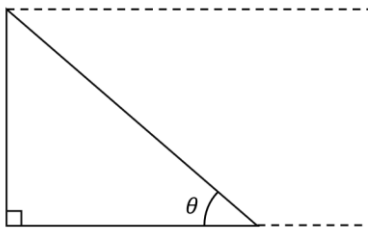
- **Line of sight** is a straight line from your eye to an object.
- Angle of **elevation**: The angle you turn to look up from the horizontal.
- Angle of **depression**: The angle you turn to look down from the horizontal.
- The **horizontal** is a line parallel to the ground.



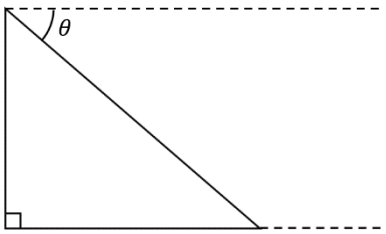
🔍 **Identify** angles of elevation and depression

Identify whether the angle shown is an angle of elevation, depression, or neither.

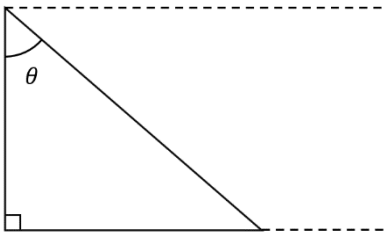
a



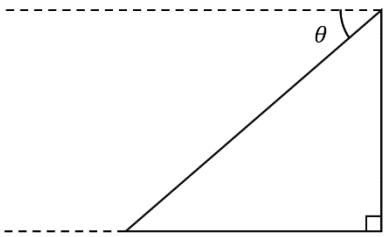
b



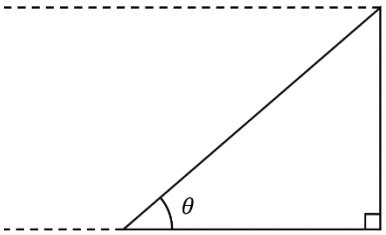
c



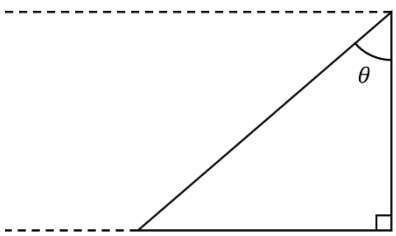
d



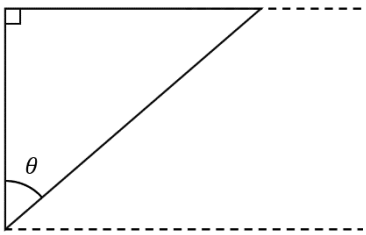
e



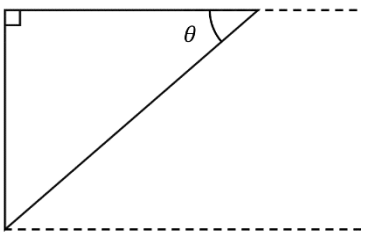
f



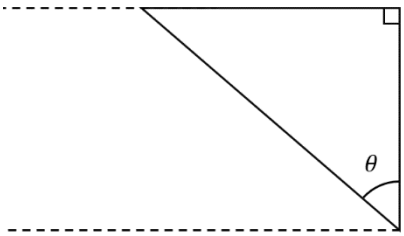
g



h



i



Finding Angles of Elevation and Depression

- To find angles, we can use these angle relationships:
  - Angles in a right-angle sum to $90°$
  - Angle sum of a triangle is $180°$
  - Alternate angles on parallel lines are equal.

Example Calculate angles of elevation and depression

Calculate the angle of elevation.

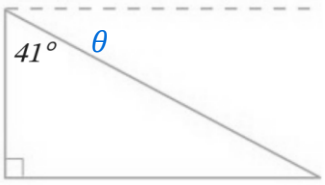


$\theta + 70° + 90° = 180°$ (angle sum of a triangle)

$\theta + 160° = 180°$

$\theta = 20°$

Calculate the angle of depression.



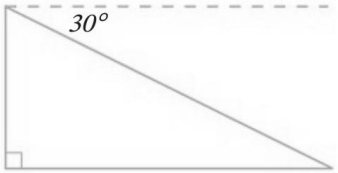
$\theta + 41° = 90°$ (angles in a right angle)

$\theta = 90° - 41°$

$\theta = 49°$

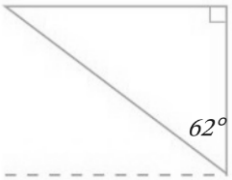
Guided Practice

a Calculate the angle of elevation.



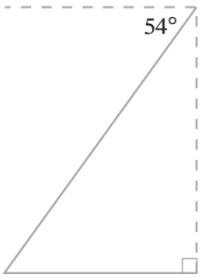
30°

b Calculate the angle of depression.



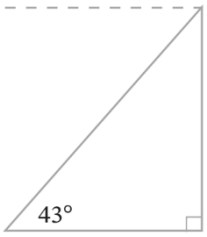
28°

c Calculate the angle of elevation.



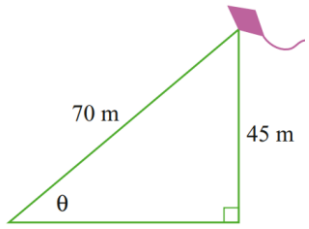
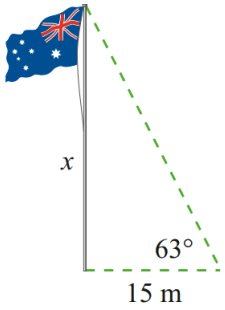
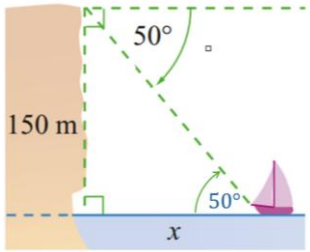
54°

d Calculate the angle of depression.



43°

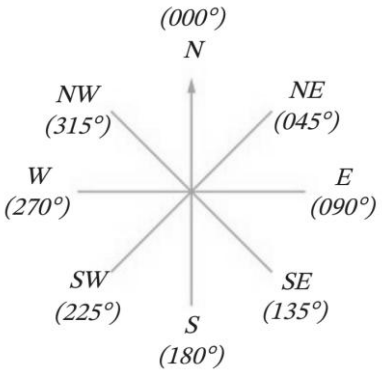
Example Solve problems involving angles of elevation and depression

Worked Example	Your Turn
A kite is flying at a height of 45 m above the ground at the end of a string of length 70 m. Find the angle of elevation from the ground to the string.  $\sin\theta = \frac{45}{70}$ $\theta = \sin^{-1}\left(\frac{45}{70}\right)$ $\theta \approx 40°$	A kite is flying at a height of 35 m above the ground at the end of a string of length 60 m. Find the angle of elevation from the ground to the string, answer to the nearest minute. 35°41′
The angle of elevation of the top of a flagpole from the ground, as observed from a point 15 m from its base, is 63°. Find the height of the flagpole.  $\tan(63°) = \frac{x}{15}$ $15\tan(63°) = x$ $x \approx 29.4\text{ m}$	The angle of elevation of the top of a flagpole from the ground, as observed 50 m from its base, is 38°. Find the height of the flagpole to 1 d.p. 39.1 m
The angle of depression from the top of a vertical cliff, 150 m above sea level, to a boat below is 50°. Find the distance of the boat from the base of the cliff to the nearest metre.  $\tan(50°) = \frac{150}{x}$ $x = \frac{150}{\tan(50°)}$ $x \approx 126\text{ m}$	The angle of depression from the top of a vertical cliff, 135 m above sea level, to a boat below is 37°. Find the distance of the boat from the base of the cliff to the nearest metre. 179 m

True Bearings

Bearings: https://www.geogebra.org/m/er3jqacn

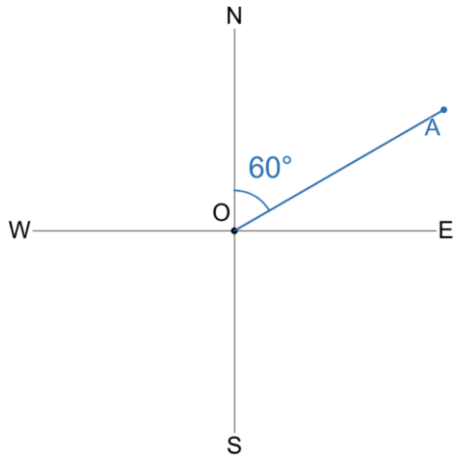
- **Bearings** give a travel direction from one point to another.
- With bearings, we are always taking a **top-down view**.



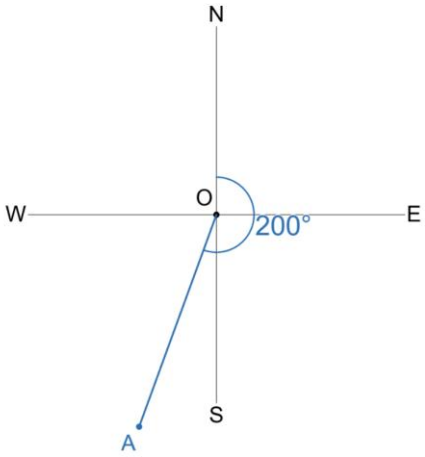
- **True bearing** measures the angle to turn clockwise from north.
- True bearings **always use three digits**.
  If a navigator only hears two digits, then they know they need it to be repeated.

Example Sketch true bearings

The bearing of A from O is 060°



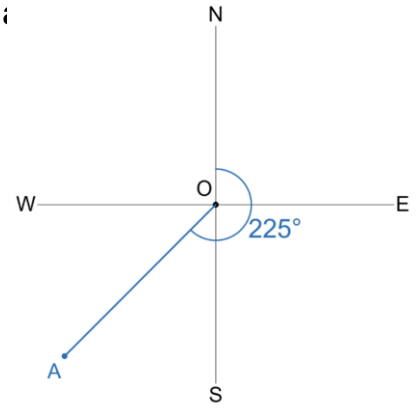
The bearing of A from O is 200°



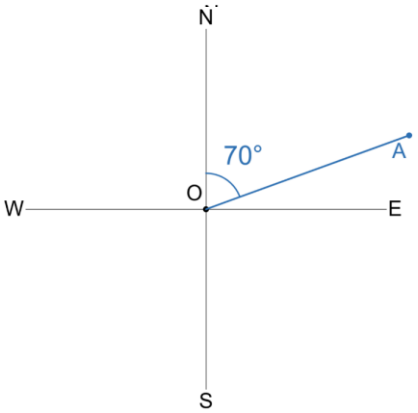
Guided Practice

Sketch these true bearings of A from O.

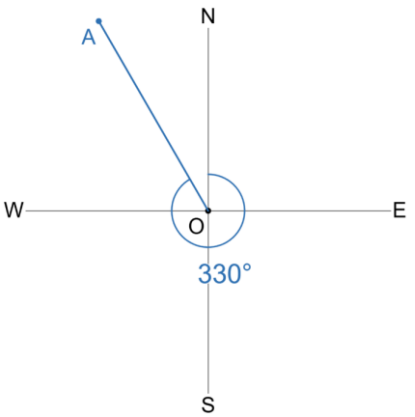
a 225°



b 070°

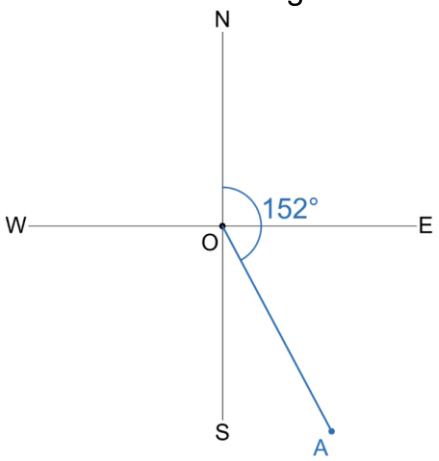


b 330°



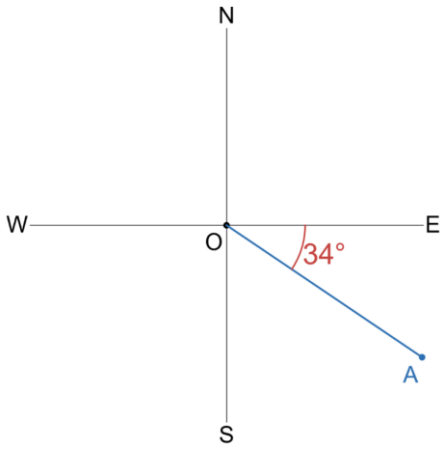
Example Interpret true bearings

Write the true bearing of A from O



$152°$

Write the true bearing of A from O



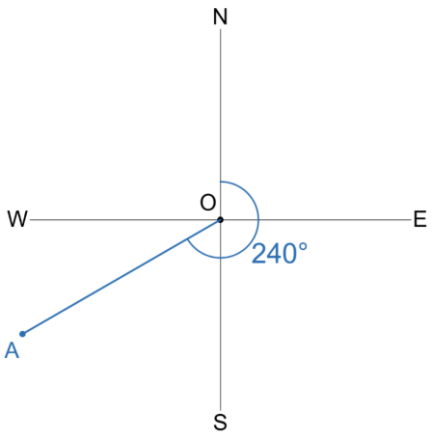
$90° + 34°$

$= 124°$

Guided Practice

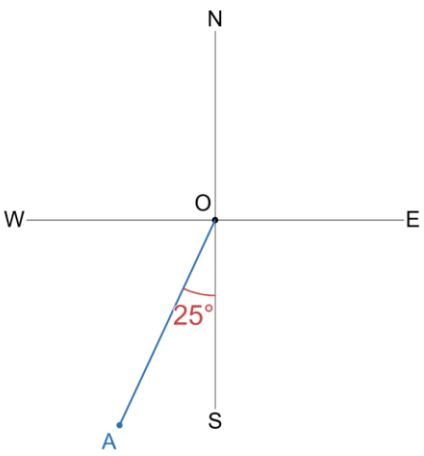
Write the true bearing of A from O

a



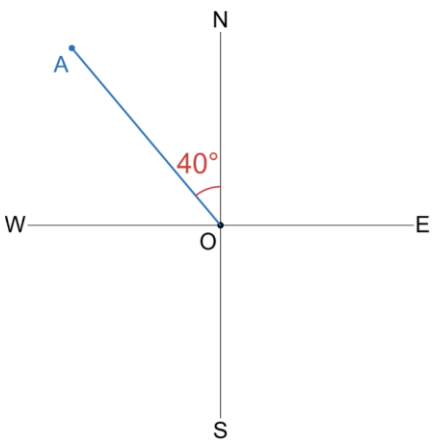
$240°$

b



$205°$

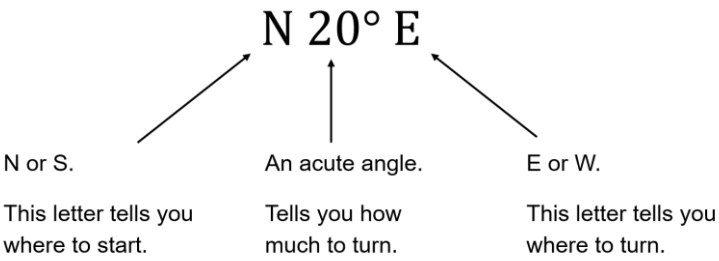
c



$320°$

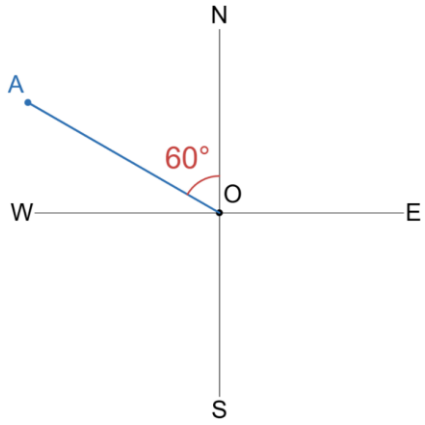
Compass Bearings

Compass bearings tell you an **acute** angle to turn, starting from **North** or **South**.

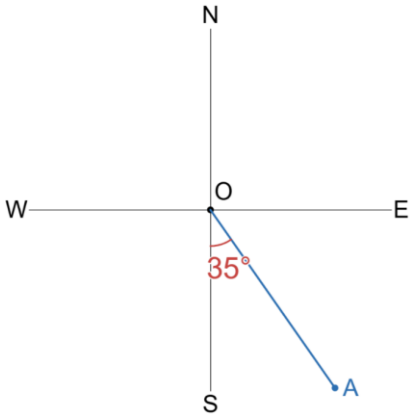


Example Sketch a compass bearing

The bearing of A from O is N60°W



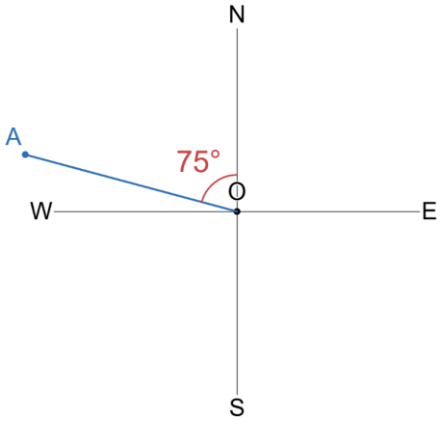
The bearing of A from O is S35°E



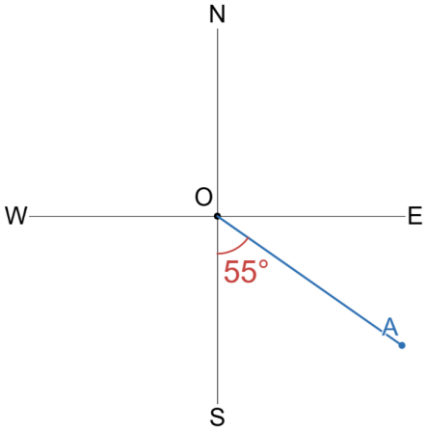
Guided Practice

Sketch these compass bearings of A from O.

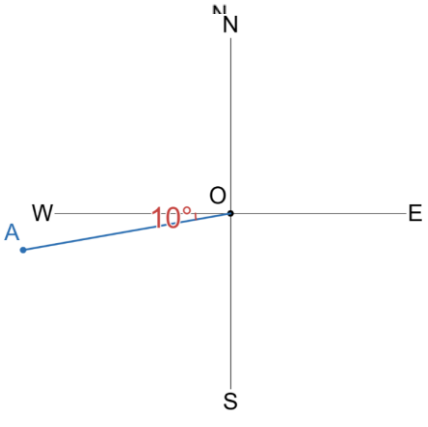
a N75°W



b S55°E



c S80°W

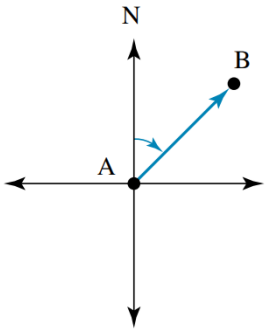


Bearing Between Two Locations

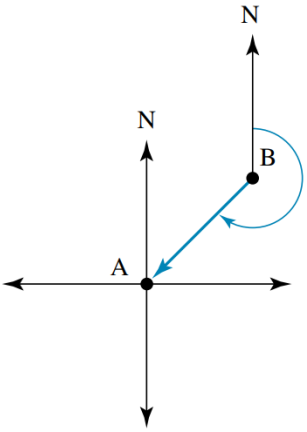
- "**From**" shows where you start a bearing.
- Bearing of "B from A" means: stand at A, find direction to B.
- In navigation, we use true bearing; compass bearings are rare.

Bearings: https://www.geogebra.org/m/rpvgcshx

B from A



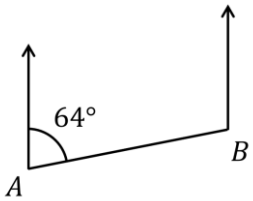
A from B



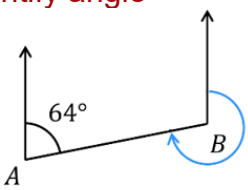
1. Identify on your diagram the angle you are trying to find.
2. Find the angle using angle relationships.

Example Determine the bearing between two locations

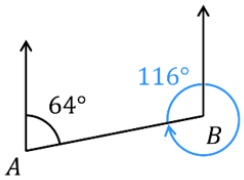
What is the bearing of A from B?



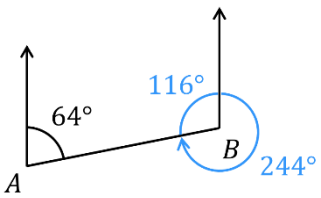
1. Identify angle



2.



Cointerior angles



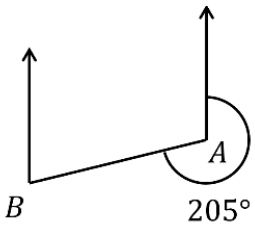
Angles at a point

The bearing of A from B is 244°

Guided Practice

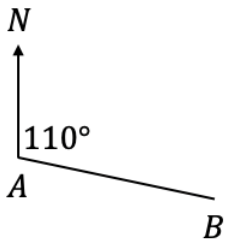
Determine the true bearing of

a A from B



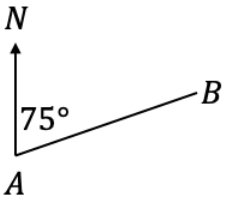
025°

b A from B



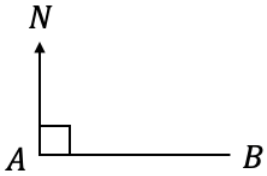
255°

a A from B



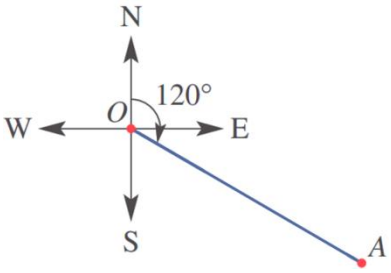
255°

b A from B



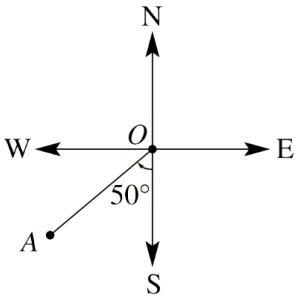
270°

c O from A



300°

d O from A



050°

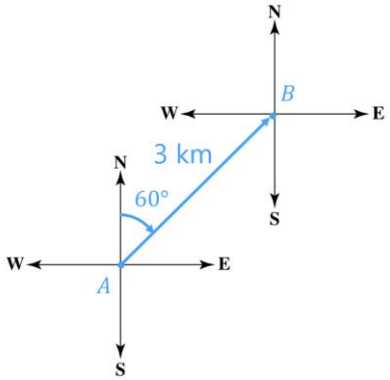
Problems involving Bearings and Trigonometry

1. Draw a diagram and a compass rose at each point.
   Label all known information (known angles, distances, what you are trying to find).
2. Draw an appropriate right-angled triangle.
3. Find the unknown using trigonometry

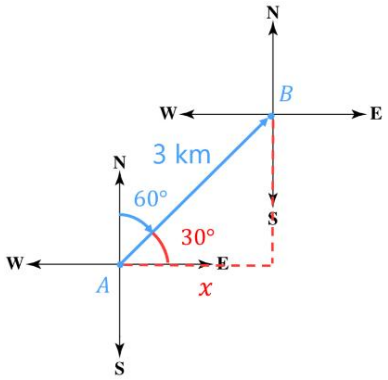
Example Solve problems involving bearings and trigonometry

A bushwalker walks 3 km on a bearing of 060° from point A to point B.
Find how far east (correct to 1 decimal place) point B is from point A.

Step 1: Diagram



Step 2: Triangle



Step 3: Solve

$$\cos(30°) = \frac{x}{3}$$

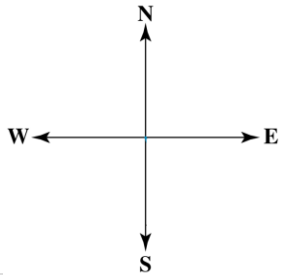
$$3\cos(30°) = x$$

$$2.6 \text{ km} = x$$

The bushwalker is 3 km east of where they started.

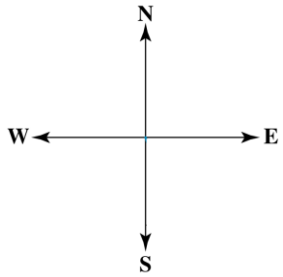
Guided Practice

a A fishing boat starts from point O and sails 75 km on a bearing of 160° to point B.
How far east of its starting point is the boat, to the nearest kilometre?



26 km

b An athlete runs 10 km on a bearing of 200° from the start to the finish line.
How far south does the triathlete run? Answer to the nearest kilometre.

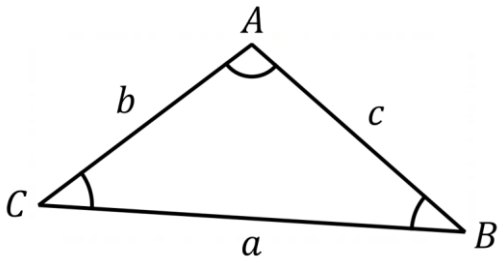


9 km

Sine Rule

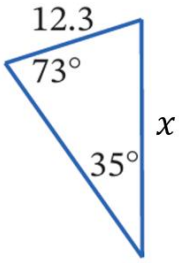
ⓘ Sine Rule (Finding a Side)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



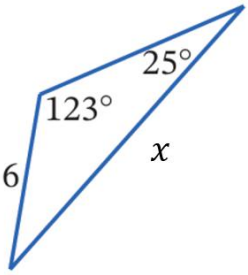
🔍 **Write** sine rule equation from a diagram

a

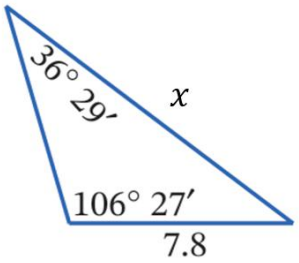


$$\frac{x}{\sin 73°} = \frac{12.3}{\sin 35°}$$

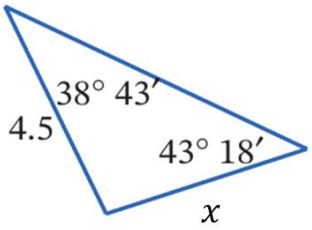
b



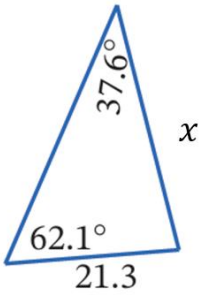
c



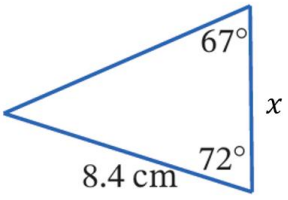
d



e

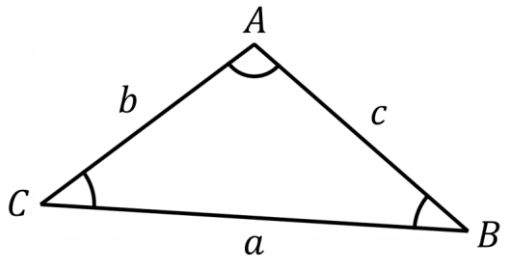


f



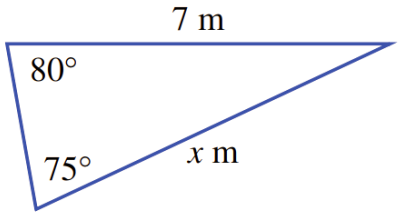
Sine Rule (Finding a Side)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Example Calculate a side length using sine rule

Find the length of the missing side, to 1 d.p.



$$\frac{x}{\sin 80°} = \frac{7}{\sin 75°}$$

$$\times \sin 80° \quad \times \sin 80°$$

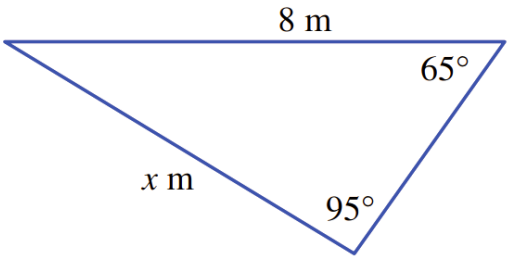
$$x = \frac{7 \sin 80°}{\sin 75°}$$

$$\approx 7.1 \text{ m}$$

Guided Practice

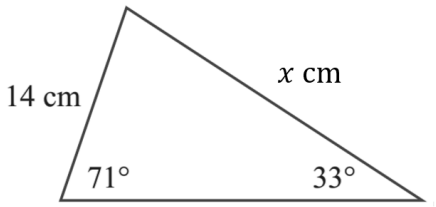
Find the length of the missing side, to 1 d.p.

c



7.3 m

d



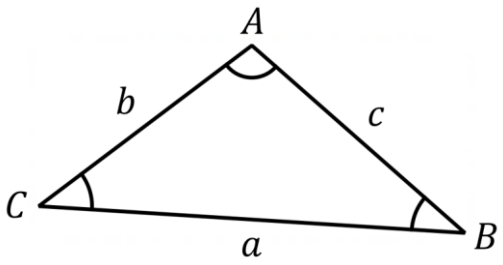
24.3 cm

Sine Rule (Finding an Angle)

When finding an angle, have sin in the numerator.

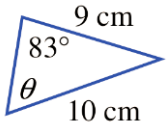
This makes it easier to solve for an angle.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



Example Calculate an angle using sine rule

Find the missing angle to the nearest minute.



$$\frac{\sin\theta}{9} = \frac{\sin 83°}{10}$$

$$\times 9 \quad \times 9$$

$$\sin\theta = \frac{9\sin 83°}{10}$$

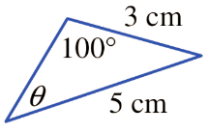
$$\theta = \sin^{-1}\left(\frac{9\sin 83°}{10}\right)$$

$$\approx 63°17'$$

Guided Practice

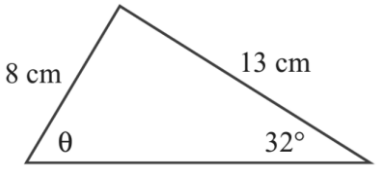
Find the missing angle to the nearest minute.

a



36° 13'

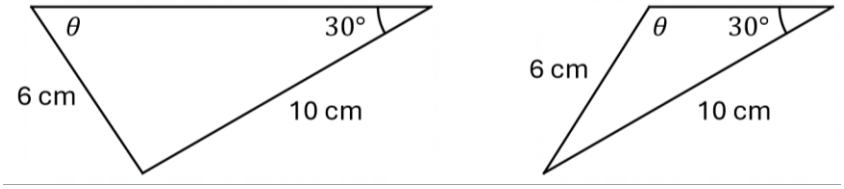
b



59° 27'

Sine Rule (Finding an Obtuse Angle)

Given two sides and one angle, it is sometimes possible to make 2 triangles:



This is due to the identity: $\sin\theta = \sin(180° - \theta)$

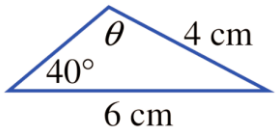
The calculator will always give the acute angle when using $\sin^{-1}(\quad)$

To get the obtuse angle, **subtract the result from 180**: $180° - \theta$

You will learn the reason for this if you study Mathematics Advanced

Example Calculate an angle using sine rule

Find the missing angle to the nearest degree, given it is obtuse.



$$\frac{\sin\theta}{6} = \frac{\sin 40°}{4}$$

$$\times 6 \quad \times 6$$

$$\sin\theta = \frac{6\sin 40°}{4}$$

$$\theta = \sin^{-1}\left(\frac{6\sin 40°}{4}\right)$$

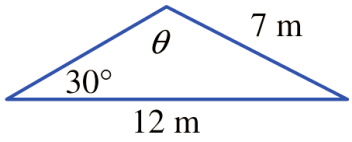
$$\approx 75°$$

$$\therefore \theta = 180° - 75° = 105°$$

Guided Practice

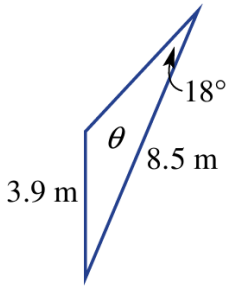
Find the missing angle to the nearest degree, given it is obtuse.

a



121°

b

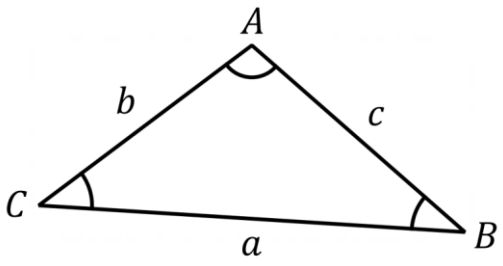


138°

Cosine Rule

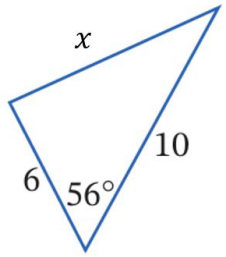
ⓘ Cosine Rule (Finding a Side)

$$c^2 = a^2 + b^2 - 2ab\cos C$$



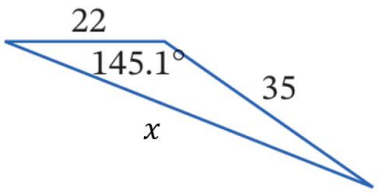
🔍 **Write** Cosine rule equation from a diagram

a

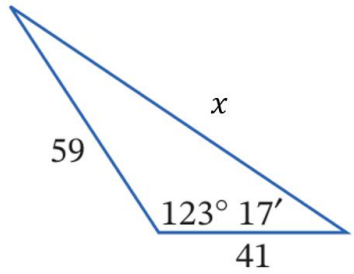


$x^2 = 6^2 + 10^2 - 2(6)(10)\cos 56°$

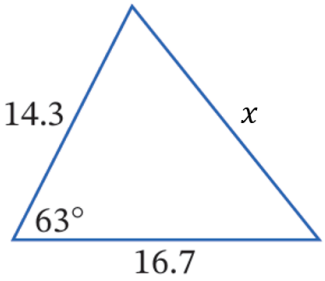
b



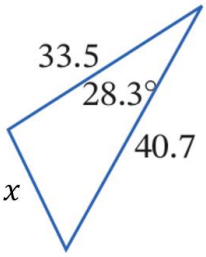
c



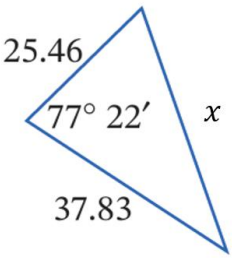
d



e

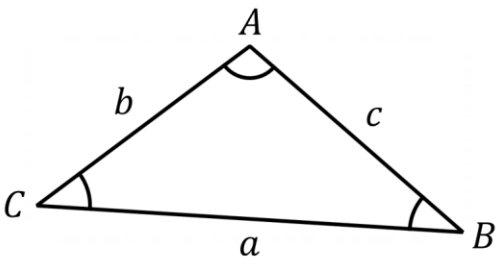


f



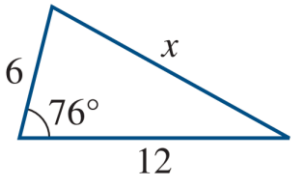
Cosine Rule (Finding a Side)

$$c^2 = a^2 + b^2 - 2ab\cos C$$



Example Calculate a side using cosine rule

Calculate the length of x to 2 d.p.

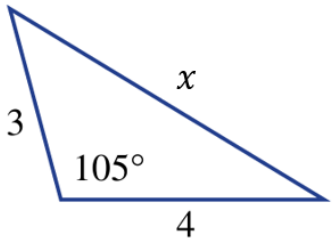


$$c^2 = a^2 + b^2 - 2ab\cos C$$
$$x^2 = 6^2 + 12^2 - 2(6)(12)\cos 76°$$
$$x = \sqrt{6^2 + 12^2 - 2(6)(12)\cos 76°}$$
$$\approx 12.05$$

Guided Practice

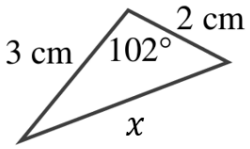
Calculate the length of x to 2 d.p.

a



5.59

b



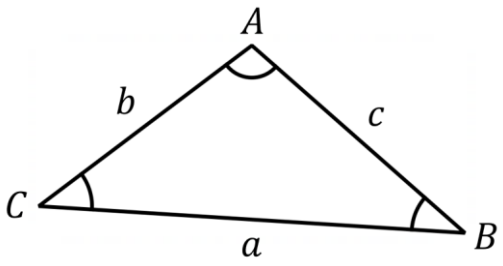
3.94

ⓘ Cosine Rule (Finding an Angle)

c is always the side OPPOSITE the angle you want to find.

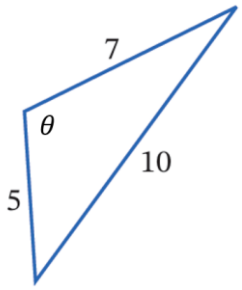
The other two sides are a and b (order doesn't matter).

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



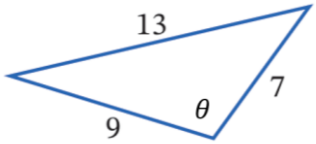
🔍 Write Cosine rule equation from a diagram

a

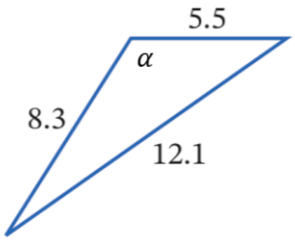


$$\cos\theta = \frac{5^2 + 7^2 - 10^2}{2(5)(7)}$$

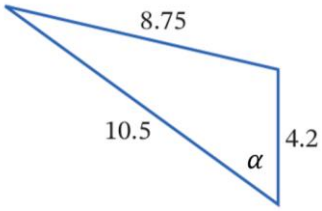
b



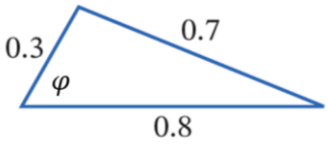
c



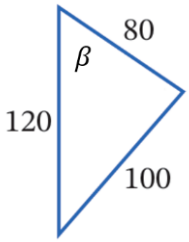
d



e

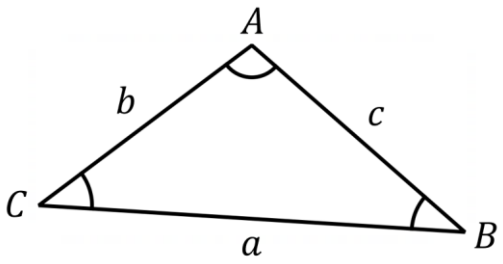


f



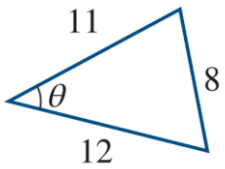
Cosine Rule (Finding an Angle)

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



Example Calculate an angle using cosine rule

Calculate θ to the nearest minute.



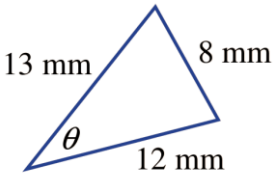
$$\cos\theta = \frac{11^2 + 12^2 - 8^2}{2(11)(12)}$$

$$\theta = \cos^{-1}\left(\frac{11^2 + 12^2 - 8^2}{2(11)(12)}\right)$$

$$\theta = 40°25'$$

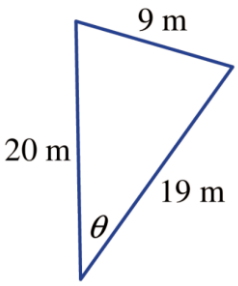
Guided Practice

a



37° 3′

b

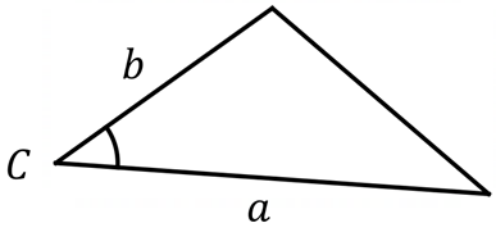


26° 32′

Area of a Triangle

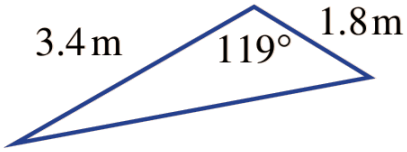
SAS Area Rule

$$A = \frac{1}{2}ab\sin C$$



Example Calculate area of a triangle using SAS area rule

Find the area of this triangle, to 1 d.p.

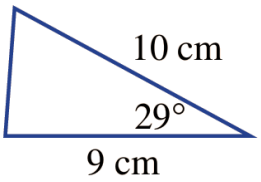


$$A = \frac{1}{2}ab\sin C$$
$$= \frac{1}{2}(3.4)(1.8)\sin 119°$$
$$\approx 2.7 \text{ km}^2$$

Guided Practice

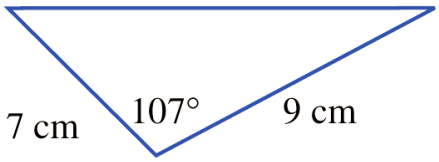
Find the area of these triangles, to 1 d.p.

a



21.8 cm²

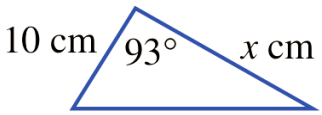
b



30.1 cm²

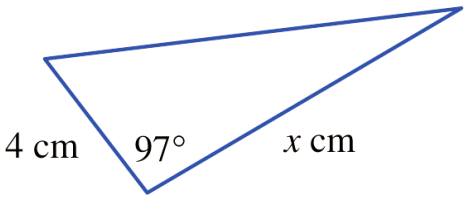
Find the side length, to 2 d.p., given the area.

a $A = 70$ cm²



14.02 cm

b $A = 37$ cm²



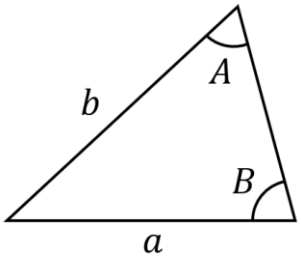
18.64 cm

Selecting an Appropriate Method and Mixed Problems

ⓘ Methods for Non-Right-Angled Triangles

For non-right-angled triangles, determine what is involved in the question:

2 sides 2 angles

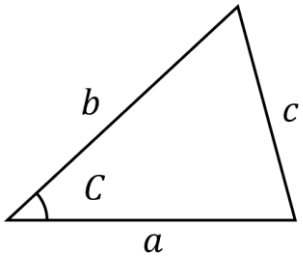


Sine Rule

$$\frac{\sin A}{a} = \frac{\sin B}{B}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

3 sides 1 angle

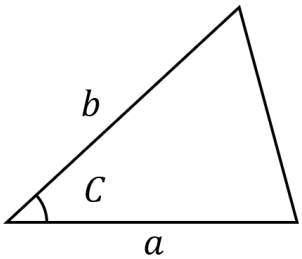


Cosine Rule

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Area

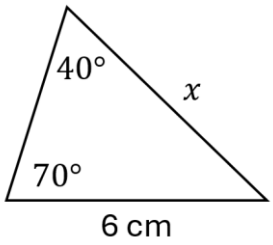


SAS Area Rule

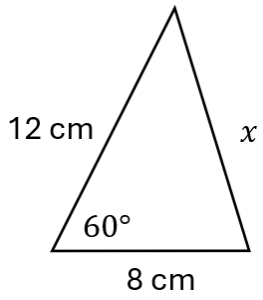
$$A = \frac{1}{2}ab\sin C$$

🔍 **Identify** which rule you would use to find the unknown

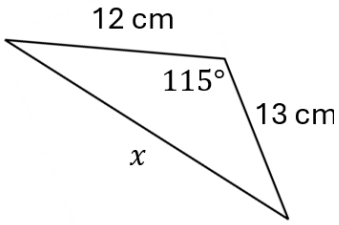
a



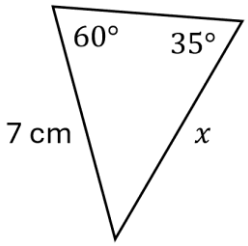
b



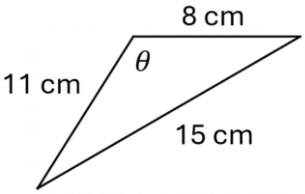
c



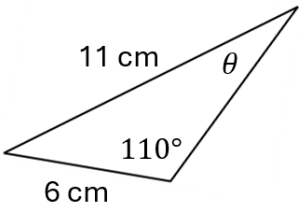
d



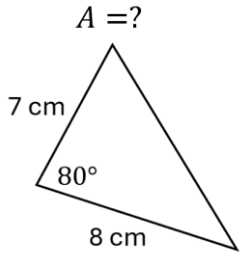
e



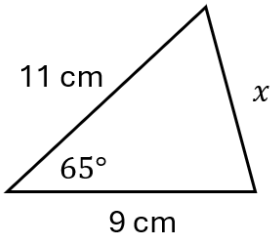
f



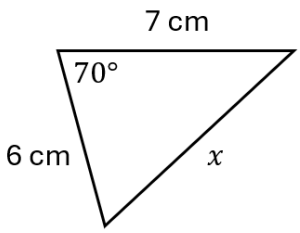
g



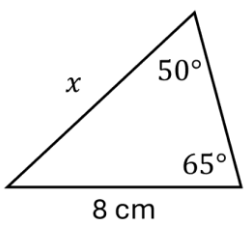
h



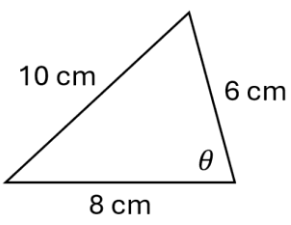
i



j



k



l

